

ZEERBOX · DESIGN PRECIS

ZeerBox design precis

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DOCUMENT CONTROL

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REVISION HISTORY

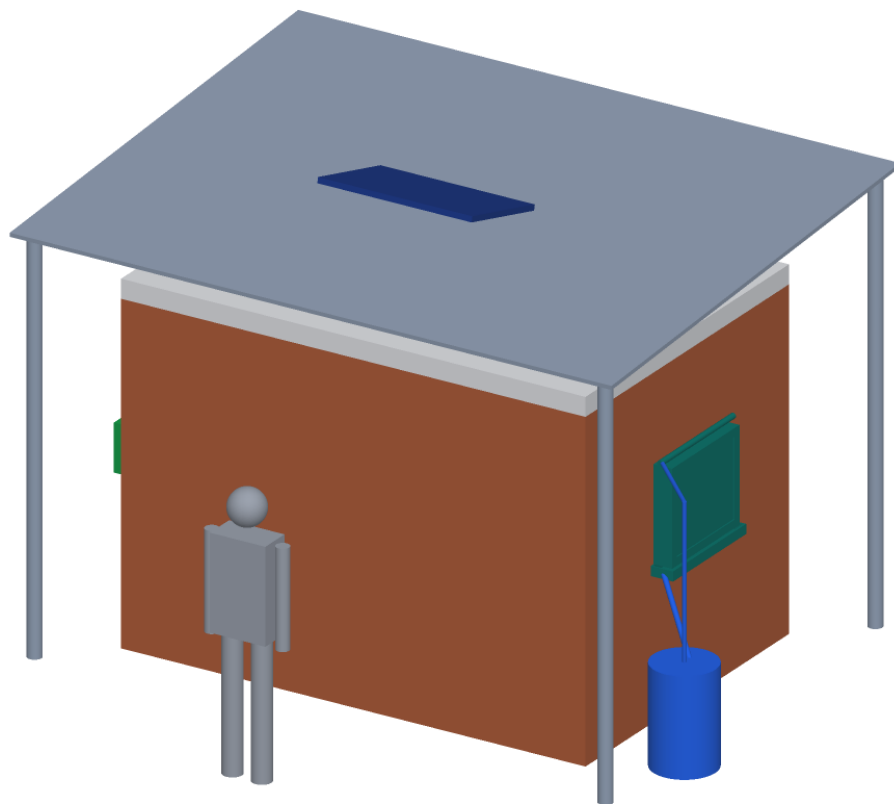
Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, control modes, first-order numbers, cost, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

ZeerBox design precis

ZeerBox is a walk-in store, 2.4 x 1.5 x 2.0 m inside, with double brick walls around a wet sand cavity, a shade roof, a 100 mm cellulose pad in the back wall, and two 12 V exhaust fans beside the door that pull outside air through the pad and along the aisle between two shelving racks. A controller reads temperature and humidity inside and outside and chooses between evaporative cooling, night ventilation and hold, and a 100 W solar panel with a small LiFePO4 battery powers it all. First-order numbers suggest that on a 35 °C, 30 % RH day the store holds about 480 kg of produce at about 26 °C, roughly 9 K below the outside air, using about 55 L of water and about 350 Wh of solar energy. The complete store costs about \$575 in parts, well over the \$300 budget; the cooling equipment alone costs about \$235.

ZeerBox

CONCEPT, NOT FOR FABRICATION



Grey: 1.75 m person for scale

Figure 1. ZeerBox massing model with the pad and sump on the back wall, solar panel on the shade roof and a 1.75 m person for scale. The door and fans are on the far end.

■ How it works

1. **Wet.** A 12 V pump in a 60 L sump drum lifts water to a perforated header on top of the pad. Water runs down through the cellulose pad and drains back to the sump through a gutter. A perforated pipe keeps the sand in the wall cavity damp, as in a zero energy cool chamber, so the walls sit near the outside wet-bulb temperature and let in less heat.

2. **Cool.** Two 250 mm fans in the front wall exhaust air from the store, which pulls outside air in through the wetted pad on the opposite wall. Evaporation cools the air from 35 °C toward its wet-bulb temperature of about 21.5 °C; with a pad effectiveness of about 75 %, it enters at about 25 °C and 74 % RH.
3. **Store.** Air sweeps along the 0.6 m aisle and across 24 crates on two three-level timber racks, picking up about 550 W of heat from the walls, the produce and the day's warm field produce, and leaves through the fans at about 27.5 °C.
4. **Decide.** A controller compares the inside and outside readings every minute and runs one of four modes (below). A three-color lamp by the door shows the mode, so users know when to keep the door shut and when the store is not cooling.
5. **Power.** A 100 W panel on the shade roof charges a 12.8 V, 12 Ah LiFePO4 battery through a PWM charge controller. The battery covers cloud and about 2 h of evening ventilation.

The shade roof sits about 450 mm above an insulated ceiling so that sun never falls on the storage room, and its overhang shades the walls for most of the day.

ZeerBox: air and water path on the design day, 35 °C and 30 % RH (all values are estimates)

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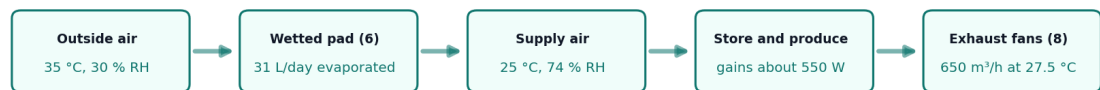


Figure 2. Air and water path on the design day, 35 °C and 30 % RH. All values are estimates.

Control modes

Mode	When (proposed thresholds)	Fans	Pump	Lamp
Evaporative cooling	Outside wet-bulb depression 4 K or more and store above its set point (proposed 20 °C)	On, speed by store temperature	On, duty cycled to keep the pad wet	Green
Night ventilation	Outside air at least 2 K cooler than the store, whatever its humidity	On	Off	Blue
Hold	Outside warmer than the store and too humid to cool	Off, or one fan at low speed for one air change per hour	Off	Amber
Guard	Store RH above 95 % for 60 min, sump float low, or battery low	Low	Off	Red, flashing

The controller logs both sensor pairs and the mode every 10 min to a memory card, so a partner can compare seasons and sites. Firmware is out of scope at TRL 2 and 3 beyond a labeled sketch of this logic.

■ Main components

Numbers match the exploded view (Figure 3) and `bom/bom.csv`.

#	Component	Proposed choice	Notes
1	Double brick walls and floor	Two 115 mm fired-brick leaves around a 75 mm cavity, 305 mm total; brick floor on compacted sand	Mud brick or block possible; proposed, awaiting Amish
2	Wet sand cavity fill	River sand kept damp by a perforated pipe along the top of the cavity	ZECC practice; about 1.5 m ³ of sand
3	Insulated ceiling	Timber boards with about 50 mm of straw, rice husk or foam and a plastic vapor sheet	Keeps roof heat out
4	Shade roof on posts	Corrugated steel on four timber or steel posts, about 3.9 x 3.0 m, 450 mm clear above the ceiling	Carries the solar panel
5	Insulated door with seal	Timber frame, plywood skins, 50 mm insulation, rubber seal, inside release	Opens from inside without a key
6	Cellulose pad, frame, header and gutter	100 mm pad, 0.6 x 0.5 m (0.3 m ²), in a frame on the back wall with a drip header and return gutter	Poultry-house or greenhouse pad
7	Sump drum, pump and hoses	60 L covered drum, 12 V submersible pump, float switch, feed and return hoses	Drained and cleaned weekly
8	Exhaust fans (pair)	Two 250 mm 12 V DC axial fans with guards and gravity shutters	About 12 W each
9	Controller and sensors	Low-cost microcontroller, two digital temperature and RH sensors (inside, outside in a radiation shield), MOSFET drivers, memory card, mode lamp, IP65 box	Generic parts
10	Solar panel	100 W monocrystalline, about 1,000 x 670 mm, on the shade roof	150 W option, see R7
11	Power box	PWM charge controller, 12.8 V 12 Ah LiFePO ₄ battery with BMS, 15 A fuse, in a ventilated box	Shaded, outside the store
12	Shelving racks	Two timber racks, three levels, 2.2 m long and 0.45 m deep	Termite-treated timber or steel angle
13	Produce crates	24 ventilated crates, about 500 x 350 x 300 mm, 20 kg each	User supplied; not in the cost

ZeerBox: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Double brick walls and floor
- 2 Wet sand cavity fill
- 3 Insulated ceiling
- 4 Shade roof on posts
- 5 Insulated door with seal
- 6 Cellulose pad, frame, header, gutter
- 7 Sump drum, 12 V pump and hoses
- 8 Exhaust fans, 12 V DC (pair)
- 9 Controller and RH/T sensors
- 10 Solar panel, 100 W
- 11 Power box: charger, LiFePO4, fuse
- 12 Shelving racks
- 13 Produce crates (user supplied)

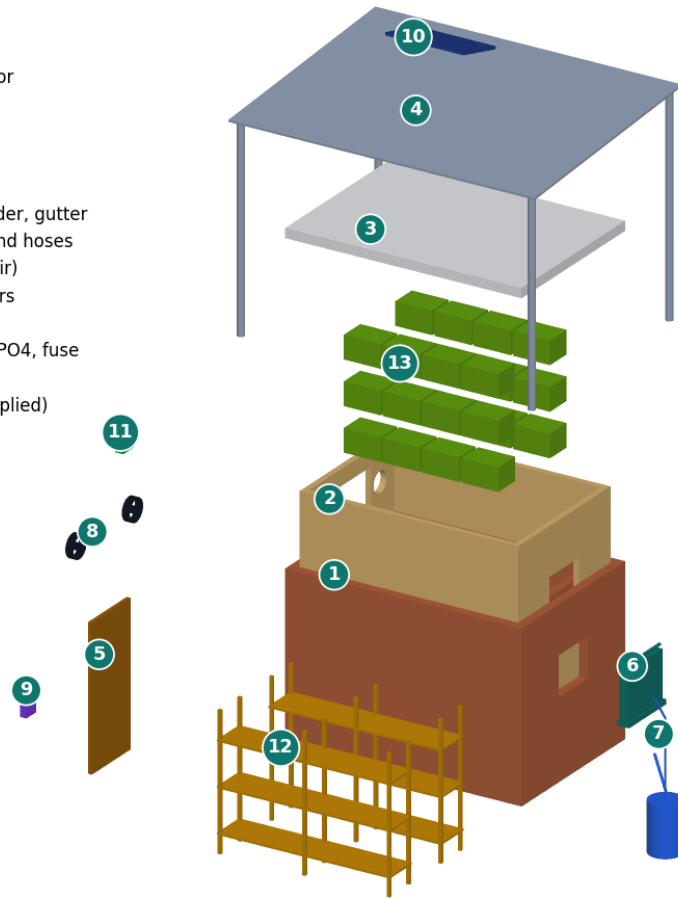


Figure 3. Exploded view with numbered callouts matching the BOM. The racks are drawn out in front; the crates, ceiling, roof and panel are lifted.

ZeerBox: cutaway

CONCEPT, NOT FOR FABRICATION

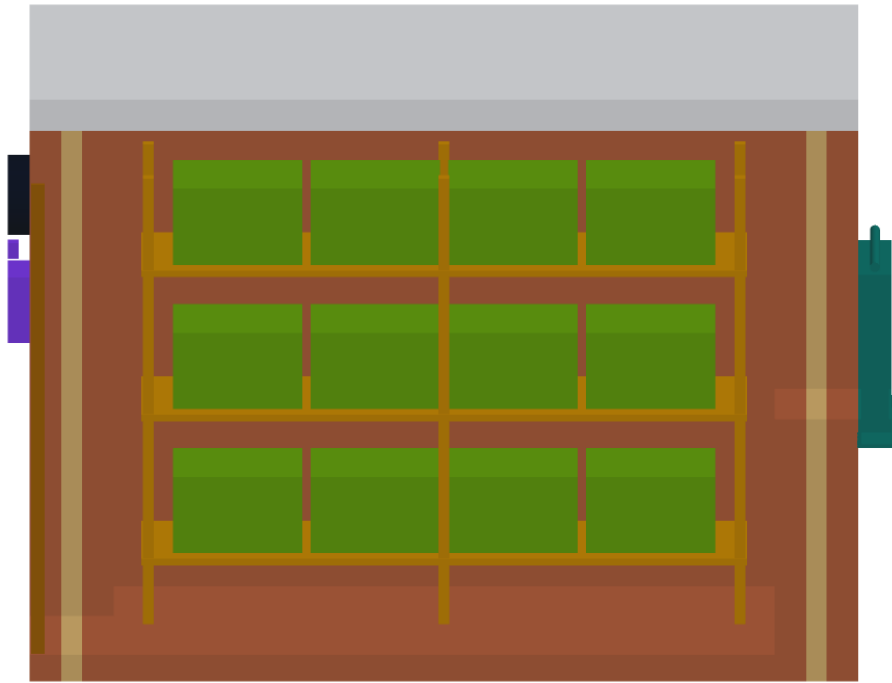


Figure 4. Section along the aisle, looking at one rack. The door and fans are at left, the pad at right, and the wet sand cavity shows as the pale band inside each end wall.

■ First-order numbers

All values are estimates for concept review and will be checked at TRL 3.

Air, pad and store temperature

Assumptions: outside air at 35 °C and 30 % RH, sea-level pressure; pad effectiveness 75 %; supply air warms by 2.6 K across the store.

Quantity	Estimate	Basis
Outside wet-bulb temperature	about 21.5 °C	Psychrometric relation at 35 °C, 30 % RH
Supply air after the pad	about 24.9 °C, 74 % RH	35 minus 0.75 x (35 minus 21.5)
Exhaust air	about 27.5 °C, about 63 % RH before produce moisture	Supply plus 2.6 K at constant moisture
Mean store air	about 26 °C, 65 to 75 % RH	Average of supply and exhaust; produce adds moisture
Temperature drop below ambient	about 9 K	R2 (8 K) met
Moisture added by the pad	about 4.2 g per kg of air	Sensible heat removed divided by latent heat

This matches field experience: D-Lab measured more than 8 °C of cooling from brick chambers in Mali's dry season. It also shows the limit. The store runs at about 26 °C, well above the 12.5 to 15 °C that tomatoes prefer, so ZeerBox slows spoilage rather than stopping it.

Heat balance and airflow

Heat gain	Estimate	Basis
Walls	about 250 W	15.6 m ² x U about 1.8 W/(m ² ·K) x 9 K, cavity taken as dry (conservative)
Ceiling	about 25 W	3.6 m ² x U about 0.8 W/(m ² ·K) x 9 K, shaded by the roof
Produce respiration	about 70 W	480 kg x about 0.15 W/kg near 26 °C, from tomato respiration data
Field heat of new produce	about 80 W	100 kg/day x 3.9 kJ/(kg·K) x 6 K, removed over 8 h
Door openings and leaks	about 50 W	Allowance
Margin	about 75 W	Allowance
Total	about 550 W	

Quantity	Estimate	Basis
Airflow needed	about 0.18 m ³ /s (about 650 m ³ /h, 380 cfm)	550 W / (1.15 kg/m ³ x 1,006 J/(kg·K) x 2.6 K)
Air changes	about 90 per hour	650 m ³ /h over 7.2 m ³
Pad face velocity	about 0.6 m/s	0.18 m ³ /s over 0.3 m ²
Fan pressure	about 30 Pa	Pad about 10 Pa, inlet, grilles and shutters about 20 Pa
Fan electrical power	about 22 to 24 W for both fans	5.5 W of air power at about 25 % fan efficiency

A wetted cavity should cut the wall gain substantially, since the outer wall surfaces run near the wet-bulb temperature, but this is not counted until it is checked at TRL 3.

Water

Quantity	Estimate	Basis	Requirement
Evaporated at the pad	about 31 L/day	0.21 kg/s of air x 4.2 g/kg for 10 h	
Wall cavity watering	about 20 L/day	ZECC practice of watering one to three times a day; allowance	
Bleed, splash and cleaning	about 4 L/day	Allowance to limit mineral build-up in the pad	
Total	about 55 L/day		R8 (70 L) met

Energy

Assumptions: fans about 24 W, pump about 6 W at full duty, controller and sensors about 1 W; 10 h of cooling and 2 h of evening ventilation on the design day; 5 peak sun hours; 30 % combined loss for panel heat, dust, PWM charging and wiring.

Quantity	Estimate	Basis	Requirement
Load while cooling	about 31 W	24 + 6 + 1	
Daily energy	about 350 Wh	31 W x 10 h plus 24 W x 2 h	
Panel yield, 100 W	about 350 Wh/day	100 W x 5 h x 0.7	R7 at risk: no margin
Panel yield, 150 W option	about 525 Wh/day	150 W x 5 h x 0.7	R7 met with 50 % margin
Battery	154 Wh, about 120 Wh usable	12.8 V x 12 Ah x 80 %	About 4 h at full load

Humid weather

Outside air	Supply air	Mean store (estimate)	Drop	Mode
35 °C, 30 % RH (design point)	about 24.9 °C	about 26 °C	about 9 K	Evaporative cooling
38 °C, 15 % RH	about 23.9 °C	about 25 °C	about 13 K	Evaporative cooling
32 °C, 50 % RH	about 25.7 °C	about 27 °C	about 5 K	Evaporative cooling
30 °C, 75 % RH	about 27.2 °C	about 28.5 °C	about 1.5 K	Hold by day, night ventilation

In the rainy season ZeerBox does little more than a shaded, ventilated shed. The controller's job then is to avoid making things worse: no wet pad adding humidity to a damp room, and cool night air let in whenever it is available.

Cost

Group	Indicative cost	Requirement
Structure (items 1 to 5 and 12), materials only	about \$340	
Cooling equipment kit (items 6 to 11 and 14)	about \$235	
Total, crates excluded	about \$575	R12 (\$300) not met, about 90 % over
Crates (item 13, user supplied)	about \$95 if bought	Not in the total

Labor for the mason and carpenter is not included.

■ Key design choices

Every choice below is **Proposed, awaiting Amish**.

- **Forced air through a pad, not a passive chamber.** A fan and pad give a larger, walk-in store with shelves at working height and a controlled airflow; a passive ZECC is cheaper and needs no power but holds far less and cannot respond to humidity. Recommendation: forced air, as in the pitch.
- **Exhaust fans, pad on the opposite wall (negative pressure).** Pulling air through the pad spreads it evenly over the pad face and keeps the fan motors out of the humid supply air. Blowing air in through the pad is the alternative. Recommendation: exhaust.
- **Wet sand cavity walls.** Borrowed from the ZECC to cut wall heat gain. Costs more brick and about 20 L/day of water; a single insulated wall is the alternative. Recommendation: wet cavity, to be checked by calculation at TRL 3.
- **Size.** 2.4 x 1.5 x 2.0 m inside for 24 crates. A smaller store (for example 1.6 x 1.5 m and 12 crates) would cost about a third less; a larger one would serve a group. Recommendation: keep 24 crates until the partner confirms the need.
- **Panel size.** 100 W has no margin on the design day (R7). Recommendation: 150 W, about \$20 more.
- **Battery.** A small LiFePO4 battery allows evening ventilation and rides through cloud. PV-direct operation with no battery saves about \$50 but stops the fans at dusk. Recommendation: keep the battery.
- **Set point and thresholds.** A 20 °C store set point, 4 K wet-bulb depression threshold and 95 % RH guard are starting values for field tuning.
- **Budget.** The complete store is about \$575 against \$300. Options are in `docs/REVIEW.md`. The `project.yaml` budget is unchanged.

■ Safety

Safety: ZeerBox is a walk-in room that people enter, with a lithium iron phosphate battery, spinning fans, standing warm water, heavy masonry and a roof that can catch the wind. Treat each of these as a hazard at every stage.

- **Entrapment and air quality.** The door must open from inside at all times, with no outside lock that can be closed on a person inside. Produce respire and uses oxygen; with the fans off and door shut for a long time, open the door and let the store air out before entering.
- **Water-borne bacteria.** Warm recirculating water in pads and sumps can grow *Legionella* and other bacteria, and the fine droplets from a pad can carry them. Cover the sump, drain and clean it weekly, dry the pad out regularly, never use the sump water for drinking or washing produce, and replace a pad that smells or grows slime. Covered water also stops mosquitoes breeding.
- **Mold and spoilage.** A damp, poorly ventilated store can spread mold between crates. The guard mode, weekly cleaning and removal of rotten produce are part of operation.
- **Battery.** A 154 Wh LiFePO4 battery is far less prone to thermal runaway than other lithium chemistries but can still overheat or short. Use a battery with a BMS, fuse it at the terminal, keep it shaded and ventilated outside the store, and do not charge it below 0 °C or above 45 °C.

- **Electrical.** All wiring is 12 V DC and below the touch-safety threshold, but water and wiring meet at the pad and pump. Use IP65 connections, drip loops and a fused supply.
- **Moving parts.** Fans need guards on both faces. Isolate power before cleaning a fan or the pump.
- **Structure.** Masonry walls and a roof that people stand under must be built by a competent mason. Anchor the roof posts and sheets against uplift in storms, keep the wet cavity from undermining the footings, and inspect for cracks each season.
- **Food safety.** Cooling slows spoilage; it does not make produce safe to eat. ZeerBox is not for meat, fish, milk or medicines.

■ Open questions for TRL 3

- Check the heat balance with a wet cavity, sun on the walls at low sun angles and realistic door opening, and size the fans and pad from that.
- Find a way to meet R3 (85 % RH), for example by wetting the inner leaf or cutting airflow at night, and check the effect on temperature.
- Confirm pad effectiveness and pressure drop for a 100 mm pad at about 0.6 m/s from supplier data.
- Decide panel size (R7) and whether the battery stays.
- Review MIT D-Lab's forced-air evaporative chamber work in Kenya in detail and contact the team.
- Close the cost gap or propose a budget change (R12).
- Choose the first partner, site and crop mix for co-design.

Concept media: [blueprint sheet](#), [interactive 3D model](#).